Ask What Matters

Adaptive AI for Smarter Travel Reviews

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The Problem

Reviews capture what travelers say — but miss what they don't.

67%

of sub-rating fields
are blank across reviews

7,200

reviews, most covering
the same 3–4 topics

180+

days since many fields
were last confirmed

Standard review forms ask the same generic questions regardless of what’s already known.

The Opportunity

"What information is unknown or outdated about a property, and what's the easiest way to learn it?"

Missing

No data exists — the field has never been rated or mentioned by any guest.

Stale

Data exists but may be outdated — recent reviews suggest conditions have changed.

Contradicted

The listing claims one thing, but guest reviews consistently say otherwise.

Our Solution

1–2 smart follow-up questions, tailored to each property, different every time.

BUILD Offline Enrichment

Vectorize, categorize, aggregate 7,200 reviews

ASK Real-Time Follow-ups

Score gaps, select questions, parse answers

Adaptive

Questions change based on each property's current knowledge state

Diverse

Cluster-diverse selection covers different property dimensions

Transparent

Every question shows its full scoring rationale

Low-Friction

Voice + text, max 20 words, no pleasantries

Data Foundation: Build Phase

Every review passes through a 4-step enrichment pipeline before it can inform question selection.

1

Detect Language

langdetect
Deterministic seed

→

2

Translate to English

GPT-4.1-mini
temp = 0.0

→

3

Compute Embedding

text-embedding-3-small (384d)

→

4

Tag Review Topics

28 topics
11 clusters

7,200 reviews enriched · 12 properties · 8 parallel workers · SHA-256 LLM cache for idempotent reruns

Knowledge Representation

Dual EMA Tracking

Short-term EMA

Half-life = 5 reviews

Captures recent sentiment shifts

Long-term EMA

Half-life = 30 reviews

Historical baseline for comparison

Drift Detection

When $|\text{short} - \text{long}| > 0.5$ with 5+ mentions, the field is flagged as drifting — conditions have changed.

55 Fields per Property

15 Sub-ratings

overall, cleanliness, service ...

12 Schema fields

pet policy, amenities, check-in ...

28 Guest topics

WiFi, breakfast, pool, noise ...

Each stored as a FieldState: value_known, EMA pair, mention count, last confirmed date

Four-Factor Scoring

Every field scored by a weighted composite of four independent signals.

55% Missing

Is this field completely unknown? Highest priority.

25% Stale

Has information aged past 180 days, or has EMA drift been detected?

15% Coverage Gap

Do few guests ever mention this? Inverse-frequency scoring.

20% Cross-Reference

Do the property listing and guest reviews disagree?

Weights are intentionally > 100% — factors reinforce, not partition.

```
score = 0.55 × missing + 0.25 × stale + 0.15 × coverage + 0.20 × cross_ref
```

Cluster-diverse selection: Q1 and Q2 always come from different property dimensions.

The Live Experience

11-Step Real-Time Pipeline

1. Detect language
2. Translate to English
3. Compute 384d embedding
4. Tag against 28-topic taxonomy
5. Persist review + tags to SQLite
6. Re-aggregate field states
7. Rank all fields (4-factor scoring)
8. Exclude already-covered topics
9. Select 1–2 questions (cluster-diverse)
10. Render natural-language questions
11. Detect listing contradictions

Key Features

< 3s latency
Full pipeline end-to-end

Natural phrasing
LLM generates ≤20-word questions

Voice + text input
Low friction for travelers

Smart parsing
Regex-first, LLM fallback

Feedback loop
Answers update states instantly

Full Transparency

Every question comes with a complete explanation — no black box.

“Why This Question?” — shown for every follow-up

Rank & Cluster

"Rank #3 of 55 fields · Room cluster"

Plain-English Reason

"No reviews mention WiFi speed yet"

Score Breakdown Bars

Visual bars for each of the 4 scoring factors

Raw Formula

$0.55 \times 1.00 + 0.25 \times 0.00 + 0.15 \times 0.92 + 0.20 \times 1.00$

Scoring Intermediates

Age: 42 days, drift detected, response rate: 3.5%

Review Overlap

"You mentioned: WiFi, Breakfast — excluded"

Runner-ups

#2 Bed Comfort (0.49) · #3 Noise Level (0.42)

Coverage Impact

"Answering raises coverage from 45% → 47%"

Impact & Scalability

Impact

- Self-improving: each answer sharpens the next question
- Contradiction alerts surface listing inaccuracies
- Richer property data with every single review

Scalability

- SQLite → Postgres for production workloads
- YAML-configurable taxonomy and scoring weights
- SHA-256 keyed LLM cache — near-instant reruns
- Multilingual from day one (langdetect + translation)

Thank You